

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I _M. Kyathi Sai Priya/192572101_____ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: M. Kyathi

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly	Uses relevant	Uses vague or	No useful

		relevant cloud examples to strengthen the answer	examples with minor mismatch	partially relevant examples	examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

Definition : Cloud Computing is a storage device which delivers the computing services such as Servers, database, Networking ,software analytics over the internet.

- It allows users to access resources whenever needed without owning or maintaining physical infrastructure.

Characteristics:

1. On-Demand Self-Service

- Users can obtain computing resources automatically without human interaction with the service provider.
- Resources such as virtual machines, storage, and databases can be created whenever required.

Example:

A developer launches a virtual server on AWS in a few minutes.

Traditional Computing:

Requires purchasing hardware and waiting for IT staff to install and configure it.

2. Broad Network Access

- Services are available through the Internet.
- Resources can be accessed using laptops, smartphones, tablets, and desktops.
- Users can work from anywhere.

Example:

Google Drive files can be accessed from any device connected to the Internet.

Traditional Computing:

Applications are usually available only within an organization's local network.

3. Resource Pooling

- Cloud providers combine computing resources into a shared pool.

- Multiple customers share the same infrastructure securely using multi-tenancy.
- Resources are dynamically assigned according to demand.

Example:

Many companies use the same AWS data center while their data remains isolated.

Traditional Computing:

Each organization purchases and maintains its own dedicated hardware.

4. Rapid Elasticity and Scalability

- Resources can be increased or decreased automatically depending on workload.
- Users pay only for the resources used.

Example:

An e-commerce website increases servers during a festival sale and reduces them afterward.

Traditional Computing:

Organizations must buy additional hardware even if it is needed only occasionally.

Cloud Computing vs Traditional On-Premise Computing

Characteristic	Cloud Computing	Traditional Computing
Infrastructure	Provided by cloud vendor	Owned by organization
Cost	Pay-as-you-use	High initial investment
Scalability	Automatic	Manual
Maintenance	Provider	Organization
Accessibility	Anywhere via Internet	Mostly local network
Deployment Time	Minutes	Days or weeks

2.Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution:

Cloud computing evolved over several decades through improvements in hardware, networking, operating systems, virtualization, distributed computing, and Internet technologies.

1: Hardware Evolution

Characteristics

- Large mainframe computers
- Expensive hardware
- Centralized computing
- Multiple users shared one system

Contribution

Introduced the concept of resource sharing.

2: Personal Computers (1980s)

Characteristics

- Individual computers
- Decentralized computing
- Local software installation

Contribution

Computing became affordable and widespread.

3: Networking and Client-Server Computing (1990s)

Characteristics

- LAN and WAN networks
- Client-server architecture
- Shared databases

Contribution

Enabled communication between computers.

4: Internet Evolution

Features

- World Wide Web
- HTTP
- Browsers
- Faster Internet

Contribution

Applications became accessible globally.

5: Distributed Computing

Characteristics

- Grid computing
- Cluster computing
- Parallel processing

Contribution

Improved performance and reliability.

6: Virtualization

Characteristics

- Multiple virtual machines on one physical server
- Better resource utilization
- Isolation between users

Contribution

Foundation of modern cloud infrastructure.

7: Internet Software Evolution

Technologies

- Web Applications
- Web Services
- APIs
- SOA (Service-Oriented Architecture)

Contribution

Allowed software components to communicate over the Internet.

8: Modern Cloud Platforms

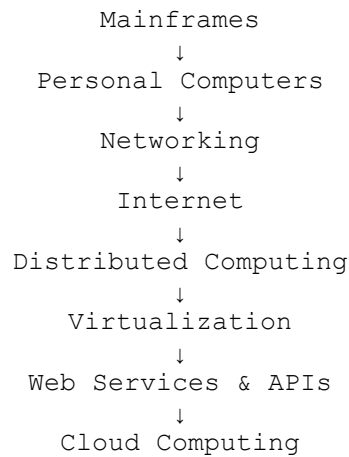
Examples:

- AWS
- Microsoft Azure
- Google Cloud

Features

- Auto scaling
- Virtual machines
- Containers
- Managed databases
- AI services
- Serverless computing

Evolution Flow



3.Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

Server virtualization allows multiple virtual machines (VMs) to run on a single physical server, improving hardware utilization and reducing costs.

What is Server Virtualization?

Server virtualization divides one physical server into several independent virtual servers.

Each VM has:

- Own operating system
- CPU allocation
- Memory
- Storage
- Applications

All VMs operate independently.

Components

- Physical Server
- Hypervisor
- Virtual Machines
- Guest Operating Systems

Role in Cloud Computing

1. Efficient Resource Utilization

Uses hardware more effectively.

2. Cost Reduction

Reduces number of physical servers.

3. Scalability

New VMs can be created quickly.

4. Isolation

Failure in one VM does not affect others.

5. Disaster Recovery

VM snapshots and backups simplify recovery.

6. Live Migration

VMs can move between physical servers without downtime.

Advantages

- Better performance
- Lower power consumption
- Easier management
- Reduced maintenance
- Improved availability

Difference Between Virtualization and Cloud Computing

Virtualization	Cloud Computing
Technology	Service Model
Creates virtual machines	Delivers services over Internet
Runs on physical hardware	Uses virtualization internally
Focus on hardware efficiency	Focus on service delivery

Virtualization

Can exist without cloud

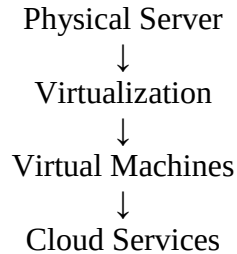
Managed locally

Cloud Computing

Depends heavily on virtualization

Managed by cloud providers

Relationship



4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.

Web services allow different applications to communicate over the Internet. The two major types are SOAP and REST.

SOAP (Simple Object Access Protocol)

Features

- Protocol
- XML-based
- High security
- Strict standards
- Uses WSDL

Advantages

- Reliable
- Secure
- Suitable for banking and enterprise applications

Disadvantages

- Complex
- Slower
- More bandwidth

REST (Representational State Transfer)

Features

- Architectural style
- Uses HTTP methods
- Lightweight
- Supports multiple data formats

HTTP Methods

- GET
- POST
- PUT
- DELETE

Advantages

- Faster
- Simple
- Easy to develop
- Ideal for mobile and cloud applications

Disadvantages

- Less standardized security
- Stateless communication

SOAP vs REST

Feature	SOAP	REST
Type	Protocol	Architectural Style
Data Format	XML only	JSON, XML, HTML
Communication	Message-based	Resource-based
Speed	Slower	Faster
Bandwidth	High	Low
Security	WS-Security	HTTPS, OAuth
Complexity	High	Low
Performance	Moderate	High

Feature**SOAP****REST**

Cloud Suitability Enterprise systems Modern cloud applications

Cloud Suitability**SOAP**

Suitable for:

- Banking
- Financial services
- Enterprise systems

REST

Suitable for:

- AWS APIs
- Google Cloud APIs
- Azure APIs
- Mobile apps
- Web applications

Conclusion

REST is preferred in cloud computing because it is lightweight, scalable, and efficient, while SOAP is preferred where high security and reliability are required.

5.Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Parameter	IaaS	PaaS	SaaS	XaaS
Full Form	Infrastructure as a Service	Platform as a Service	Software as a Service	Anything as a Service
Provides	Virtual infrastructure	Development platform	Ready-to-use software	Any cloud-based IT service
User Manages	OS, Applications, Data	Applications and Data	Only uses the software	Depends on service type
Provider Manages	Hardware, Storage, Network	Infrastructure, Runtime, OS	Entire software and infrastructure	Entire service
Target Users	System Administrators	Developers	End Users	Organizations and Businesses
Control Level	High	Medium	Low	Varies
Customization	High	Moderate	Low	Depends on service
Maintenance	User manages	Provider manages	Provider manages	Provider manages

Parameter	IaaS	PaaS	SaaS	XaaS
	software	platform	everything	service
Cost	Pay for infrastructure used	Pay for platform used	Subscription-based	Pay-as-you-use
Best For	Hosting servers and storage	Application development	Email, Office applications	AI, Security, Database, Storage services
Examples	Amazon EC2, Microsoft Azure VM	Google App Engine, Heroku	Gmail, Microsoft 365, Salesforce	AIaaS, DBaaS, STaaS, Security as a Service